

# Forecasting Dengue in Brazilian Capital Cities

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## Goal

We compare forecasting models for weekly dengue incidence in the 27 Brazilian capital cities.

- Evaluate whether climate covariates improve forecasting performance;
- Compare three model families: INLA, AR(p), and SARIMAX;
- Select the best model in each capital using mean absolute error (MAE) and weighted interval score (WIS).

### Set-up

The analysis is based on weekly dengue counts and climate covariates for each capital city.

- 27 Brazilian capital cities;
- Weekly dengue reported cases as the response;
- Climate summaries based on lagged moving averages;
- Rolling forecasting windows with horizon of 8 weeks;
- Train: 2010-2025; Test: 2025-present;
- Model comparison based on out-of-sample MAE and WIS.

### Model families

We compare three classes of models:

- **INLA models** ('M0–M7'): poisson/negbin temporal random effects with optional climate covariates;
- **AR(p)** ('M8'): autoregressive benchmark model;
- **SARIMAX models** ('M9–M13'): time-series models with exogenous climate regressors.

### Main covariates

The climate information enters through lagged summaries such as:

- temperature averaged over 8 or 12 weeks;
- minimum and maximum temperature averaged over 8 or 12 weeks;
- average maximum humidity over 12 weeks;
- average precipitation over 52 weeks.

|         | MAE        | WIS         |
|---------|------------|-------------|
| SARIMAX | 11 (40.8%) | 16 (59.26%) |
| INLA    | 10 (37.0%) | 5 (18.54%)  |
| AR(p)   | 6 (22.2%)  | 6 (22.2%)   |

Table: Number of capitals where each model wins based on each metric.

# Capital Cities Study

## National summary

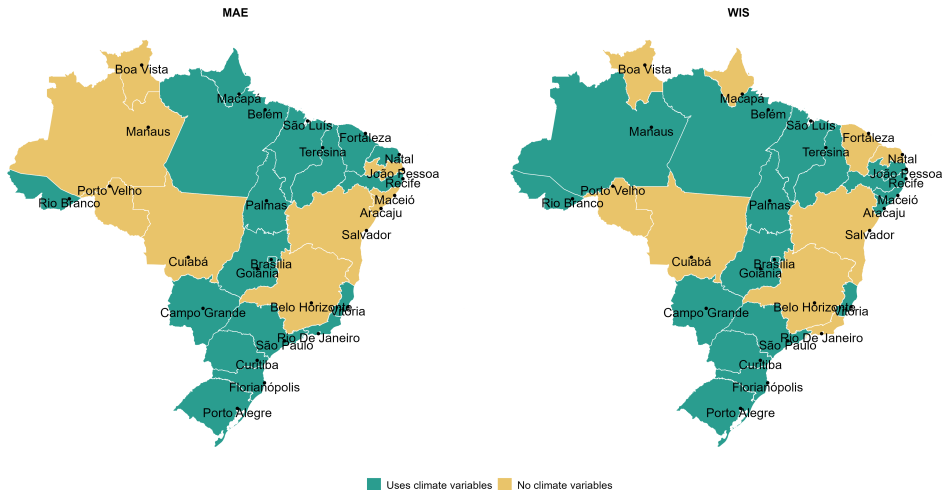


Figure: Use of climate variables comparison based on MAE and WIS.

# Capital Cities Study

## National summary

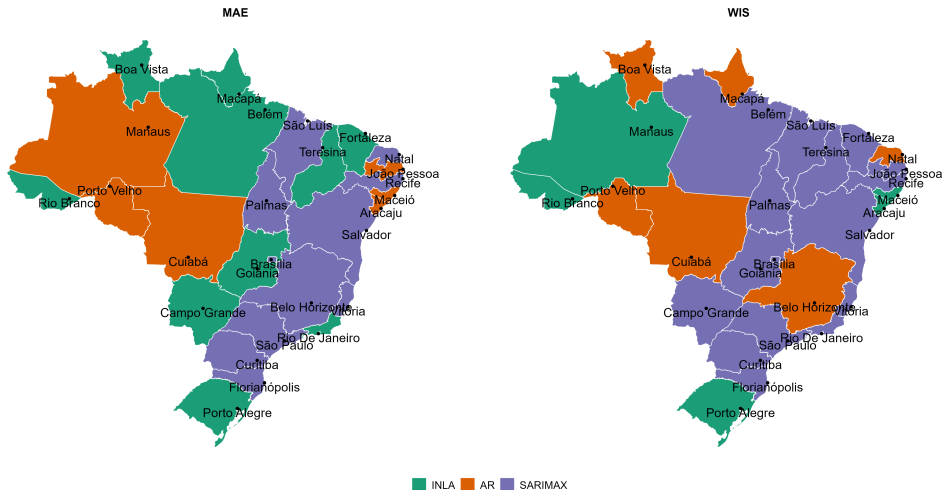


Figure: Best model type comparison based on MAE and WIS.



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Best model by capital

## SARIMAX winners

|     | MAE | WIS |
|-----|-----|-----|
| M9  | BA  | BA  |
|     | MG  | CE  |
|     |     | RJ  |
| M11 | DF  | DF  |
|     | ES  | ES  |
|     | PR  | PR  |
|     |     | GO  |
| M12 | MA  | MA  |
|     | PE  | PE  |
|     | RN  | PA  |
| M13 |     | PB  |
|     | SC  | SC  |
|     | TO  | TO  |
|     | SP  | SP  |
|     |     | MS  |
|     |     | PI  |

## INLA winners

|    | MAE | WIS |
|----|-----|-----|
| M0 | RR  |     |
| M1 | GO  |     |
|    | PA  |     |
| M3 | CE  | AL  |
|    | RJ  |     |
| M4 | MS  | AM  |
|    | PI  |     |
| M5 | AP  |     |
| M6 | AC  | AC  |
|    | RS  | RS  |
|    |     | CE  |

## AR winners

|    | MAE | WIS |
|----|-----|-----|
| M8 | AL  | RR  |
|    | AM  | AP  |
|    | MT  | MT  |
|    | PB  | MG  |
|    | RO  | RO  |
|    | SE  | RN  |

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## Forecast accuracy highlights

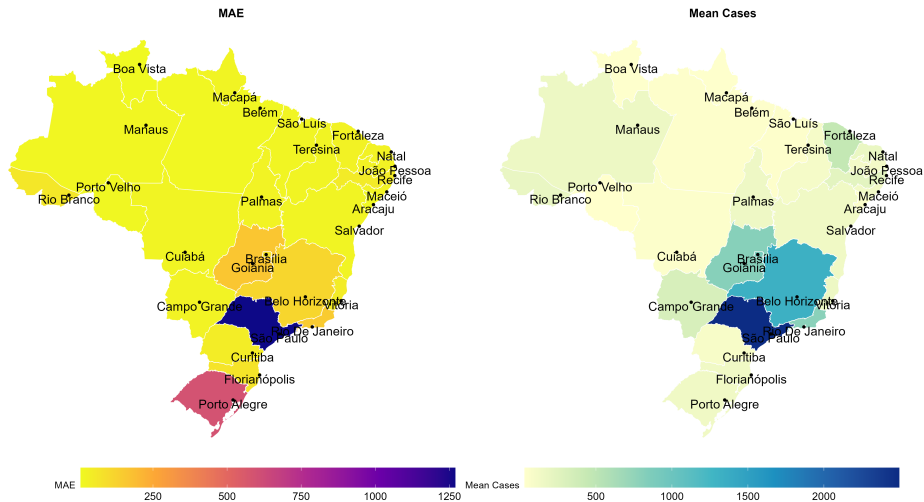


Figure: Comparison between the MAE and the averaged number of reported cases.

# Capital Cities Study

## Forecast accuracy highlights

| Lowest MAE |    |       | Highest MAE |     |         |
|------------|----|-------|-------------|-----|---------|
| RR         | M0 | 7.30  | SP          | M13 | 1268.50 |
| SE         | M8 | 9.22  | RS          | M6  | 593.22  |
| RO         | M8 | 12.71 | GO          | M1  | 164.33  |
| PA         | M1 | 14.90 | RJ          | M3  | 149.56  |

Table: Best models for the four lowest and highest MAE.

# Capital Cities Study

## Forecast accuracy highlights

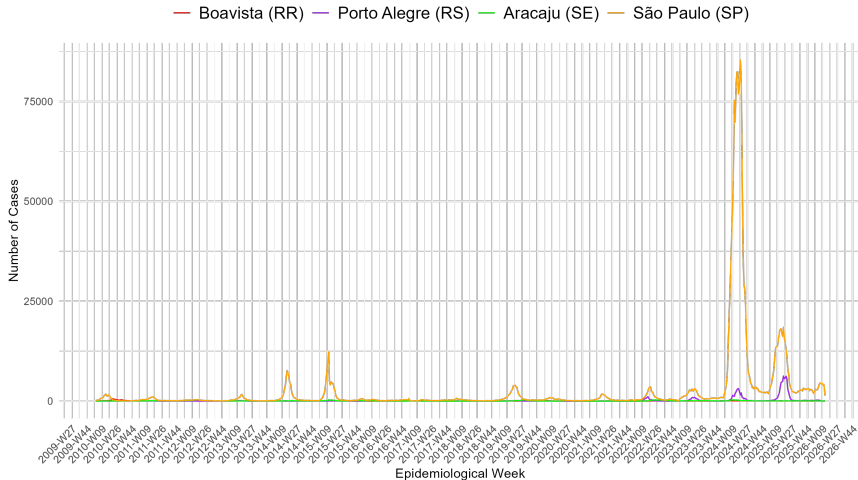


Figure: Cases time series for selected capitals.

### Take-away messages

- Based on WIS, SARIMAX seems to be a more universal best model, but in general, there is **no universal best model** across the Brazilian capitals;
- Climate information improves forecasting in all southern capitals, but there is no clear pattern in the other regions;
- Highest MAEs occur in capitals with highest number of cases/peaks.
- In several cities, WIS leads to different choices of best models compared to MAE, especially in INLA models.

## Next steps

- Implement machine learning models;
- Explore regional patterns in the winning models;

- [Bra26] Ezequiel Braga. *dengue-forecasting*.  
<https://github.com/EzequielEBS/dengue-forecasting>. 2026.

**Thank you!**